# Credit Risk Analysis

An end-to-end data analysis project using 32,409 loan applications to predict defaults, quantify portfolio risk, and deliver actionable lending recommendations — reducing expected annual losses by **$47.4M**.

CREDIT RISK PREDICTIVE MODELLING BUSINESS INTELLIGENCE

Project Scope

# 32,409 Applications. One Clear Objective.


This project analyses a comprehensive loan dataset across 12 features — from applicant demographics and loan details to credit history — to build a predictive model that quantifies risk and drives smarter approval decisions.

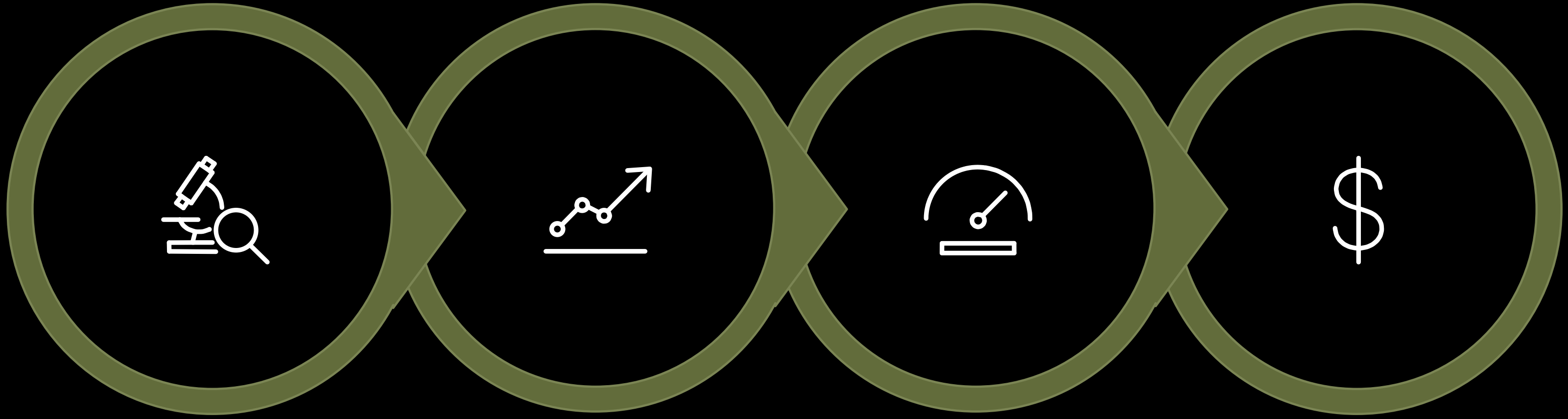
**32,409**
Loan applications analysed

**12**
Features per record

**21.87%**
Observed default rate

Methodology

# A Four-Stage Analytical Framework



**Exploratory EDA**

**Predictive Model**

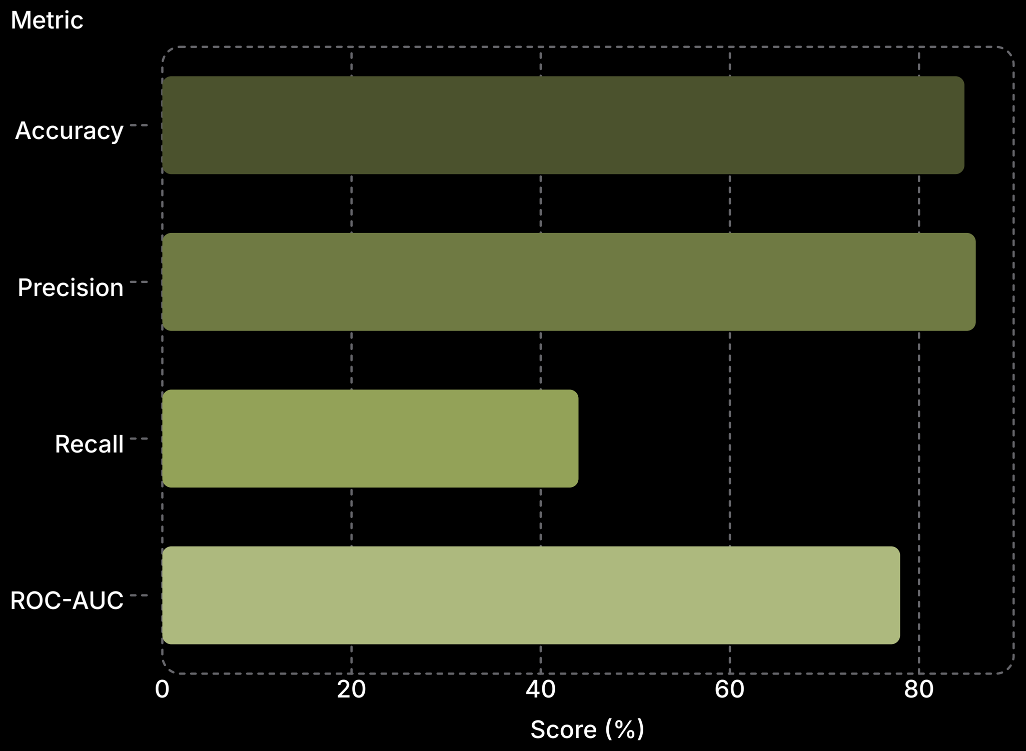
**Risk Scoring**

**Business Impact**

Each stage builds on the last — from raw data exploration through to board-ready financial recommendations — ensuring every insight is grounded in evidence and tied to business outcomes.

Model Performance

# Logistic Regression: Built for Interpretability



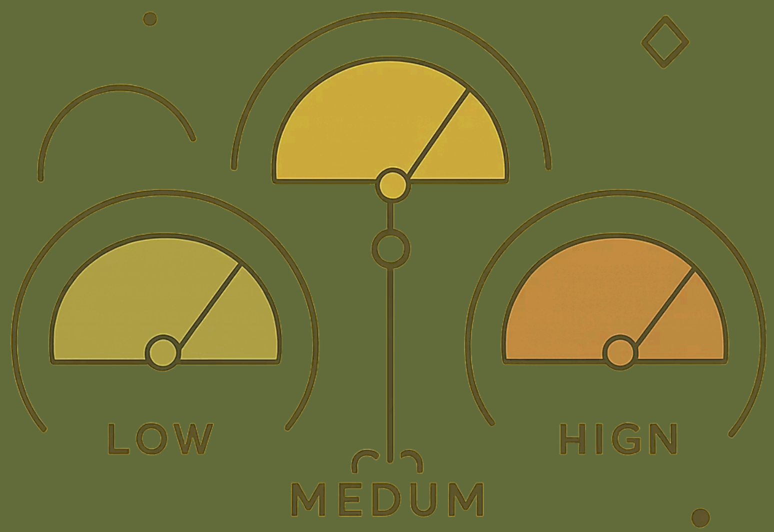

### Why These Metrics Matter

**Precision 86%** — When the model flags a safe loan, it's right 86% of the time. This protects portfolio quality.

**Recall 44%** — Intentionally conservative. The goal is not to catch every defaulter, but to avoid approving high-risk loans.

**ROC-AUC 0.78+** — Strong discriminative ability between defaulters and non-defaulters.

Logistic regression was chosen for its interpretable coefficients, probability outputs for risk scoring, and industry-standard use in credit modelling.




Risk Segmentation

# Three Tiers. Clear Decisions.

### 🟢 Low Risk (0–30)

**4.9%** default rate

14,107 applicants

Approve at base −0.5% rate

### 🟡 Medium Risk (30–60)

**15.5%** default rate

10,381 applicants

Approve with caution at base rate

### 🔴 High Risk (60–100)

**60.5%** default rate

7,921 applicants

Reject or price at base +2.5%

Key Findings

# What Drives Default Risk?

## ⚠️ Highest-Risk Indicators

### Loan Grade F–G

98.4% default rate — strongest single predictor

### Previous Default

37.9% default rate — credit history is critical

### Renting Home

31.6% default rate vs. 7.5% for homeowners

## ✅ Protective Factors

### Home Ownership

7.5% default rate — strongest protective signal

### Loan Grade A

10.0% default rate — low-risk borrowers

### Higher Income

Consistent inverse correlation with default

Business Impact

# $47.4M in Annual Savings — By Approving Fewer Loans


## The Recommended Strategy

By rejecting only the **High Risk tier** (**29.4% of applications**), the organisation retains 70.6% of its loan book while cutting expected losses by **$47.4M annually**.

**$68.0M ⤚ $20.6M**

Expected annual loss

**70.6%**

Approval rate maintained

Implementation

# From Insight to Action: A Phased Rollout

1. **0–30 Days**

   Deploy risk scoring, update underwriting matrix, adjust interest rate pricing by tier

2. **1–3 Months**

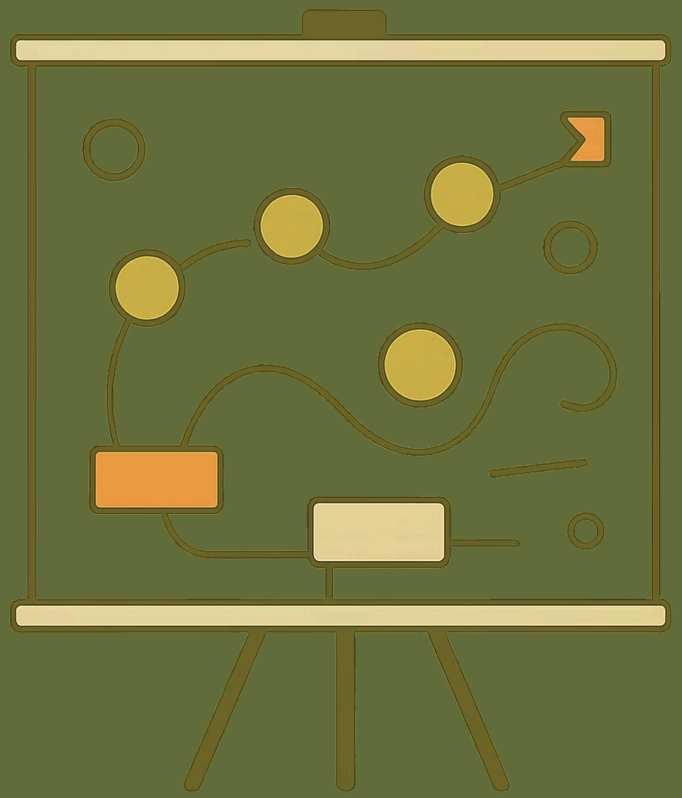
   Monitor actual vs. predicted defaults, flag borderline cases, implement automated risk alerts

3. **3–6 Months**

   Increase loss reserves, retrain model with new data, conduct portfolio concentration analysis

4. **6–12 Months**

   Build early warning system, develop intervention strategies, integrate into portfolio management



Technical Stack

# Tools That Power the Analysis

## Python

Pandas, NumPy, Scikit-learn for data manipulation, modelling, and evaluation

## Tableau

Executive Overview and Credit Risk Analysis dashboards for stakeholder communication

## Jupyter Notebook

50+ reproducible steps with markdown explanations and inline visualisations

## Logistic Regression

Interpretable coefficients, probability outputs, and industry-standard credit modelling

Project Summary

# What This Project Delivers

## Skills Demonstrated

- **Exploratory Data Analysis** — Statistical profiling across 12 dimensions
- **Predictive Modelling** — Logistic regression with ROC-AUC 0.78+
- **Risk Scoring** — 0–100 scale with three actionable tiers
- **Financial Impact** — $47.4M annual savings quantified and justified